The Blessing of Dimensionality for Policy Gradients: Sparse Recovery, Evidence Weighting, and the Logarithmic Cost of Action Space Expansion

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Abstract

The REINFORCE gradient estimator suffers from variance that appears to scale linearly with the action space dimension $d = |\mathcal{A}|$. We show this is a curse of dimensionality illusion: when the optimal policy is k-sparse (concentrates on $k \ll d$ actions), high-dimensional concentration phenomena reduce the effective sample complexity to $O(k \log(d/k))$ — *sub-linear* in d. We formalize this through three mechanisms: (1) measure concentration bounds showing gradient noise self-averages across irrelevant action dimensions; (2) compressed sensing theory giving sparse gradient recovery guarantees; (3) an effective-dimension refinement of the evidence weight from the Ω-framework that captures the true information content of the agent's experience. Combined with L1 policy regularization, this yields a natural curriculum: start sparse (low variance, fast convergence), expand as evidence improves (logarithmic cost). The blessing of dimensionality transforms the action space expansion problem from multiplicative to logarithmic.

1 The Variance Problem in Large Action Spaces

1.1 REINFORCE and the Apparent Curse

The REINFORCE gradient estimator for a policy π_θ over action space $\mathcal{A}$ with $|\mathcal{A}| = d$ is:

$$\hat{g}(\tau) = R(\tau) \cdot \nabla_\theta \log \pi_\theta(\tau) \in \mathbb{R}^d. \tag{1}$$

The variance of this estimator decomposes by action:

$$\mathrm{Var}(\hat{g}) = \sum_{a=1}^{d} \underbrace{\pi(a)}_{\text{sampling prob.}} \cdot \underbrace{\left(\frac{\nabla_\theta \pi(a)}{\pi(a)}\right)^2}_{\text{score}^2} \cdot \underbrace{R(a)^2}_{\text{reward}^2} - \|\nabla J\|^2. \tag{2}$$

For a low-probability action a with $\pi(a) = \epsilon \to 0$:

$$\text{contribution}_a \;\propto\; \frac{R(a)^2}{\pi(a)} \;\to\; \infty. \tag{3}$$

This is the **inverse probability weighting** problem: rare actions contribute enormous variance spikes when sampled. Expanding $\mathcal{A}$ by adding many low-probability actions appears to make this sum diverge.

The apparent conclusion: variance scales as $\Theta(d)$, so REINFORCE is impractical for large action spaces.

We show this conclusion is wrong when the optimal policy is sparse.

1.2 The Overfitting Analogy

The parallel to classical ML overfitting is precise:

	Supervised learning	**Policy gradient**
Space	Parameter space $\mathbb{R}^d$	Action space $\mathcal{A}$, $\lvert\mathcal{A}\rvert = d$
Problem	Too many params vs. data	Too many actions vs. samples
Noise	Fitting noise in labels	Gradient spikes from rare actions
Signal	True function (low-dim)	True gradient (sparse)
Cure	L1/Lasso (sparsity)	L1 on policy (action sparsity)
Blessing	Compressed sensing	Sparse gradient recovery

The key insight from compressed sensing: *sparsity makes high-dimensional problems easier, not harder.*

2 The Blessing: Concentration and Sparse Recovery

2.1 Concentration of the Gradient Vector

Theorem 2.1 (Gradient concentration in high dimensions)**.** *Let $\hat{g}_1, \ldots, \hat{g}_M$ be i.i.d. REINFORCE gradient samples in $\mathbb{R}^d$. Suppose the gradient covariance $\Sigma = Cov(\hat{g})$ has eigenvalues $\lambda_1 \geq \cdots \geq \lambda_d$. Then:*

$$\mathbb{P}\left(\left\|\frac{1}{M}\sum_{m=1}^{M}\hat{g}_m - \nabla J\right\| \leq \epsilon\right) \geq 1 - \delta \tag{4}$$

provided $M \geq C \cdot \frac{\mathrm{tr}(\Sigma)}{\epsilon^2} \cdot \log(d/\delta)$, where C is a universal constant.

Proof sketch. Apply the matrix Bernstein inequality to the sum of centered random vectors $\hat{g}_m - \nabla J$. The key quantity is $\mathrm{tr}(\Sigma)$ (not $d \cdot \lambda_{\max}$) because the noise contributions from different eigenspaces are independent and add their variances, not their maxima. □

Remark 2.2 (Why $\mathrm{tr}(\Sigma)$, not $d \cdot \sigma^2$)**.** *The naive bound treats all d dimensions equally: $M \geq d \cdot \sigma^2_{\max}/\epsilon^2$. But $\mathrm{tr}(\Sigma) = \sum_i \lambda_i$ can be* much *smaller than $d \cdot \lambda_1$ when the eigenspectrum decays. This is the first blessing: gradient variance concentrates on a low-dimensional subspace.*

2.2 Effective Dimension

Definition 2.3 (Effective dimension of the gradient)**.** *The effective dimension of the policy gradient at π_θ is:*

$$d_{eff} = \frac{\mathrm{tr}(\Sigma)}{\lambda_{\max}(\Sigma)} = \frac{\sum_{i=1}^{d}\lambda_i}{\lambda_1}. \tag{5}$$

This measures how many eigenvalues contribute meaningfully. If only k eigenvalues are large and the rest are negligible: $d_{\text{eff}} \approx k$. The sample complexity (**??**) becomes:

$$M \geq C \cdot \frac{d_{\text{eff}} \cdot \lambda_1}{\epsilon^2} \cdot \log(d/\delta) \approx O\left(\frac{k}{\epsilon^2}\log d\right). \tag{6}$$

This is sub-linear in d. Adding 10,000 low-probability actions to a 10-action game costs an extra $\log(10000/10) \approx 7$ factor, not a $1000\times$ factor.

2.3 Sparse Policy Recovery via Compressed Sensing

When the optimal policy π^* is k-sparse (at most k actions have positive probability in each state), compressed sensing theory gives sharp guarantees.

Theorem 2.4 (Sparse gradient recovery)**.** *Let π^* be k-sparse. Under the restricted isometry property (RIP) on the gradient measurement matrix, the true gradient direction can be recovered from M REINFORCE samples with:*

$$\|\hat{\nabla} J - \nabla J\| \leq C_1 \cdot \frac{\sigma_k(\nabla J)_1}{\sqrt{k}} + C_2 \cdot \frac{\epsilon}{\sqrt{M}} \tag{7}$$

where $\sigma_k(\nabla J)_1$ is the ℓ_1 error of the best k-sparse approximation, provided:

$$M \;\geq\; C \cdot k \log(d/k). \tag{8}$$

Proof sketch. The REINFORCE samples $\hat{g}_m$ are random linear measurements of ∇J corrupted by noise. When π^* is k-sparse, ∇J is approximately k-sparse (gradient is zero in directions corresponding to zero-probability actions). Apply the Candès–Tao RIP theory to the measurement ensemble. □

Remark 2.5 (The $k \log(d/k)$ scaling)**.** *This is the compressed sensing miracle: you need samples proportional to the sparsity k, not the ambient dimension d. The* $\log(d/k)$ *factor is the information-theoretic price of not knowing* which k *actions are relevant.*

2.4 Johnson–Lindenstrauss for Gradient Updates

Proposition 2.6 (Random projection preserves gradient structure)**.** *Let $\Phi \in \mathbb{R}^{m \times d}$ be a random Gaussian matrix with $m = O(k \log d/\epsilon^2)$. Then with high probability, for all k-sparse gradient vectors g, g':*

$$(1-\epsilon)\|g - g'\| \leq \|\Phi g - \Phi g'\| \leq (1+\epsilon)\|g - g'\|. \tag{9}$$

Implementation: Project $\hat{g}_m \in \mathbb{R}^d$ to $\tilde{g}_m = \Phi \hat{g}_m \in \mathbb{R}^m$, average in the low-dimensional space, then recover via sparse recovery (L1 minimization / LASSO). This reduces the per-step computational cost from $O(d)$ to $O(m) = O(k \log d)$.

3 Evidence Weighting with Effective Dimension

3.1 Refined Evidence Weight

The EW-PG framework (companion paper) uses the evidence weight:

$$w_i = \frac{V_{\min}}{V_i}, \qquad V_i = \mathrm{Var}(\hat{g}_i). \tag{10}$$

This treats the variance as a scalar, summing over all d dimensions. But the blessing of dimensionality reveals that most of this variance is irrelevant noise in unused action dimensions.

Definition 3.1 (Effective evidence weight)**.** *The effective evidence weight refines V_i using the gradient's effective dimension:*

$$V_i^{\mathit{eff}} = \frac{d_{\mathit{eff},i}}{d} \cdot V_i, \qquad w_i^{\mathit{eff}} = \frac{V_{\min}^{\mathit{eff}}}{V_i^{\mathit{eff}}}. \tag{11}$$

Theorem 3.2 (Effective evidence improvement)**.** *For an agent with k-sparse optimal policy in a d-action game:*

$$\frac{w_i^{eff}}{w_i} = \frac{d}{d_{eff,i}} \geq \frac{d}{k}. \tag{12}$$

The effective evidence weight is a factor of d/k larger than the ambient weight. For $d = 10{,}000$, $k = 5$: the effective weight is $2{,}000\times$ larger.

Remark 3.3 (What this means)**.** *An agent in a 10,000-action game with a 5-sparse policy has been* systematically underweighted *by the ambient evidence measure. Most of its "variance" is noise in action dimensions it will never use. The effective weight corrects this, allowing the agent to take larger steps and converge faster.*

3.2 Estimating Effective Dimension

Algorithm 1: Effective Dimension Estimation

Input: Gradient samples $\hat{g}_1, \ldots, \hat{g}_M \in \mathbb{R}^d$
Input: Threshold $\tau \in (0, 1)$ (e.g., $\tau = 0.95$)
Output: Effective dimension $\hat{d}_{\text{eff}}$

Compute sample covariance $\hat{\Sigma} = \frac{1}{M} \sum_m (\hat{g}_m - \bar{g})(\hat{g}_m - \bar{g})^\top$;

`// Option A: Ratio estimator (cheap, $O(Md)$)`

$\hat{d}_{\text{eff}} \leftarrow \text{tr}(\hat{\Sigma})/\lambda_{\max}(\hat{\Sigma})$;

`// $\lambda_{\max}$ via power iteration, $O(Md)$`

`// Option B: Eigenvalue counting (precise, $O(d^2M + d^3)$)`

Compute eigenvalues $\hat{\lambda}_1 \geq \cdots \geq \hat{\lambda}_d$;

$\hat{d}_{\text{eff}} \leftarrow \min\{k : \sum_{i=1}^k \hat{\lambda}_i \geq \tau \cdot \text{tr}(\hat{\Sigma})\}$;

`// Option C: Random projection (scalable, $O(Mm)$)`

Project: $\tilde{g}_m = \Phi \hat{g}_m$ with $\Phi \in \mathbb{R}^{m \times d}$, $m \ll d$;

$\hat{d}_{\text{eff}} \leftarrow \text{tr}(\tilde{\Sigma})/\lambda_{\max}(\tilde{\Sigma})$ in the projected space;

4 L1-Regularized Evidence-Weighted PG

4.1 The Sparse EW-PG Update

Combining L1 regularization (for sparsity) with evidence weighting (for adaptive step sizes) and the effective dimension (for accurate evidence):

Algorithm 2: Sparse Evidence-Weighted PG (Sparse-EW-PG)

Input: Game $\mathcal{G}$, N agents, $|\mathcal{A}_i| = d_i$
Input: Learning rates γ_n, L1 coefficient $\mu > 0$
Input: Sparsity relaxation schedule $\mu_n = \mu/(1 + n/n_0)$
Output: Sparse policies converging to Nash

Initialize $\pi_i \in \Pi$ for each agent;
for $n = 1, 2, \ldots$ **do**
  `// --- Gradient and evidence ---`
  for $i = 1, \ldots, N$ **do**
    Compute $\hat{v}_{i,n}$ (REINFORCE);
    Update gradient covariance estimate $\hat{\Sigma}_{i,n}$;
    Estimate effective dimension $d_{\text{eff},i,n}$ (Algorithm **??**);
    Compute effective evidence weight:;
    $V_{i,n}^{\text{eff}} \leftarrow (d_{\text{eff},i,n}/d_i) \cdot \hat{V}_{i,n}$;
    $w_{i,n}^{\text{eff}} \leftarrow V_{\min,n}^{\text{eff}}/V_{i,n}^{\text{eff}}$;
  `// --- Sparse update with proximal step ---`
  for $i = 1, \ldots, N$ **do**
    Gradient step:;
    $\tilde{\pi}_i \leftarrow \pi_{i,n} + \gamma_n \cdot w_{i,n}^{\text{eff}} \cdot \hat{v}_{i,n}$;
    L1 proximal step (soft-thresholding):;
    for *each action a* **do**
      $\tilde{\pi}_i(a) \leftarrow \text{sign}(\tilde{\pi}_i(a)) \cdot \max(|\tilde{\pi}_i(a)| - \gamma_n \mu_n,\ 0)$;
    Project back to simplex:;
    $\pi_{i,n+1} \leftarrow \text{proj}_\Delta(\tilde{\pi}_i)$;
  `// --- Adaptive sparsity relaxation ---`
  `// As evidence improves, relax L1 to allow exploration`
  `// $\mu_n$ decreases: start sparse, expand action support`

4.2 The Natural Curriculum

The interaction between L1 regularization and evidence weighting creates a **self-organizing curriculum**:

1. **Early training** (n small, μ_n large, V_i high):
   - L1 drives policy toward extreme sparsity ($k \approx 1$–2)
   - $d_{\text{eff}} \approx k \ll d \rightarrow$ effective evidence is high
   - High $w_i^{\text{eff}} \rightarrow$ large steps $\rightarrow$ fast convergence to sparse optimum
2. **Mid training** (μ_n decreasing, V_i decreasing):
   - L1 relaxes $\rightarrow$ policy support grows (k increases)
   - But evidence is improving (more data) $\rightarrow V_i$ decreases
   - Net effect: d_{eff} grows but V_i^{eff} stays controlled
   - Agent explores new actions with calibrated confidence
3. **Late training** ($\mu_n \approx 0$, V_i small):

- No sparsity constraint → full action support available
- High evidence → agent knows its policy well
- Converges to true (possibly dense) Nash equilibrium

Remark 4.1 (Comparison with entropy regularization)**.** *Standard entropy regularization* $-\beta H(\pi)$ *encourages* uniform *exploration (spread mass). L1 encourages* focused *exploration (concentrate mass). In high dimensions, L1 is strictly better:*

- *Entropy:* $d_{\text{eff}} \approx d$ *(all dimensions active)* → *no blessing*
- *L1:* $d_{\text{eff}} \approx k$ → *full blessing of dimensionality*

This is why entropy regularization helps with exploration but does not *address the variance problem, while L1 addresses both.*

5 Convergence Analysis

5.1 Main Result

Theorem 5.1 (Convergence of Sparse-EW-PG)**.** *Let* π^* *be a* k*-sparse stable Nash policy in a stochastic game with* $|\mathcal{A}| = d$*. Under the SOS condition with parameter* μ*, and with:*

- $\gamma_n = \gamma/(n+m)^p$, $p \in (1/2, 1]$
- $\mu_n = \mu_0/(1 + n/n_0)$ *(L1 annealing)*
- *Evidence weights computed with effective dimension*

the Sparse-EW-PG iterates satisfy:

$$\mathbb{E}[\|\pi_n - \pi^*\|^2] \;=\; O\left(\frac{\mathrm{HM}(V^{\text{eff}})}{\mathrm{AM}(V^{\text{eff}})} \cdot \frac{C_k}{n^q}\right) \tag{13}$$

where $C_k = O(k \log(d/k))$ *is the* sparse constant*, sub-linear in* d*.*

Proof sketch. Three steps:

1. The L1 proximal step ensures π_n stays approximately k-sparse for all n (up to the annealing schedule). Standard proximal gradient convergence gives $\|\pi_n - \pi_k^*\|_1 \to 0$ where π_k^* is the best k-sparse approximation to π^*.
2. With k-sparse iterates, the effective dimension satisfies $d_{\text{eff},n} \leq 2k$ (the gradient covariance has at most $2k$ significant eigenvalues — k from the policy support and k from the exploration directions).
3. Apply the EW-PG convergence theorem with effective variance: $V^{\text{eff}} = (d_{\text{eff}}/d) \cdot V$. The constant C_k arises from **??**: the sparse recovery bound replaces the ambient-dimension bound.

□

5.2 Comparison of Sample Complexities

Method	**Samples to ϵ-Nash**	**Scaling in d**	**Scaling in k**
Standard PG	$O(d^2/\epsilon^2)$	d^2 (quadratic)	—
PG + entropy reg.	$O(d/\epsilon^2)$	d (linear)	—
PG + L1 reg.	$O(k \log(d/k)/\epsilon^2)$	$\log d$ (logarithmic)	k (linear)
EW-PG + L1 (ours)	$O(\frac{\mathrm{HM}}{\mathrm{AM}} k \log(d/k)/\epsilon^2)$	$\log d$	$k \cdot \frac{\mathrm{HM}}{\mathrm{AM}}$

The combined method achieves *logarithmic* scaling in the action space dimension and an additional HM / AM ≤ 1 improvement from evidence weighting.

6 The Communication Connection

6.1 Sparse Policies Are Communicable

The cooperative PG (companion paper) shows that communication quality is bounded by self-knowledge, which is bounded by evidence. The blessing of dimensionality adds a crucial refinement: *sparse policies are inherently more communicable.*

To communicate a k-sparse policy over a d-action space:

- Naive: transmit d probabilities $\to$ rate $O(d)$
- Sparse: transmit k (action, probability) pairs $\to$ rate $O(k \log d)$

The $\log d$ factor (identifying which actions) replaces the d factor (transmitting all probabilities). This means:

Corollary 6.1 (Sparsity enables coalition formation)**.** *In a game with $|\mathcal{A}| = d$ and k-sparse optimal policies, the communication bandwidth required for effective coalition coordination scales as $O(k \log d)$, not $O(d)$. The self-knowledge bottleneck (§2 of Cooperative PG) is relaxed by a factor of $d/(k \log d)$.*

6.2 Vibing Is Dense; Articulation Is Sparse

The "vibing" spectrum from the cooperative paper acquires a precise meaning:

- **Vibing agent**: policy is effectively dense (spreading mass over many actions through imprecise habits). $d_{\text{eff}} \approx d$. Poor self-knowledge, poor communication.
- **Articulate agent**: policy is sparse (concentrating on the actions that matter). $d_{\text{eff}} \approx k \ll d$. Good self-knowledge, good communication.

L1 regularization drives the agent from vibing toward articulation. The blessing of dimensionality ensures this transition is *computationally cheap* in the action space dimension.

7 Connection to the Ω-Framework

7.1 The Embedding Space Interpretation

In the Ω-framework (theos §I), each agent's observation space $\mathcal{O}$ is embedded in a high-dimensional space where the blessing of dimensionality operates. The manuscript (Chapter 9) describes this as "Maximus the Confessor's logoi — every entity has a position in semantic space," and notes that "in high dimensions, structure becomes MORE visible, not less."

The policy gradient result makes this precise:

- The policy $\pi \in \mathbb{R}^d$ lives in a high-dimensional space
- The *effective* policy lives on a k-dimensional submanifold
- The gradient concentrates on this submanifold (blessing)
- The evidence weight measures information content on the submanifold
- Sparse recovery lets us find the submanifold from few samples

This is the "blessing of dimensionality" from Chapter 9 of the theological manuscript, now formalized as a convergence guarantee.

7.2 Pearl's Hierarchy and Sparse Representations

At Pearl Level 1 (association), the agent observes correlations across all d actions — a dense, high-variance representation. At Pearl Level 3 (counterfactual), the agent has identified the k *causal* actions — a sparse, low-variance representation.

The progression up Pearl's hierarchy IS the sparsification process:

$$\begin{array}{c} \text{Level 1: dense policy } (d_{\text{eff}} \approx d, \text{ high } V, \text{ “vibing”}) \\ \downarrow \quad \text{L1 regularization + evidence accumulation} \\ \text{Level 3: sparse policy } (d_{\text{eff}} \approx k, \text{ low } V^{\text{eff}}, \text{ “articulate”}) \end{array}$$

The L1 coefficient μ_n controls the speed of this ascent.

8 Numerical Illustration

8.1 Expected Behavior

For a game with $|\mathcal{A}| = 1000$ and $k = 5$ optimal actions:

Phase	d_{eff}	V^{eff}/V	**Effective samples**
Early ($n < 100$)	$\sim$3–5	$\sim$0.5%	200$\times$ more informative
Mid ($100 < n < 1000$)	$\sim$5–15	$\sim$1.5%	65$\times$ more informative
Late ($n > 1000$)	$\sim$5–30	$\sim$3%	33$\times$ more informative

The "effective samples" column shows how much more informative each sample is compared to the ambient-dimension estimate. Even in the late phase, the effective dimension is $30 \ll 1000$, giving a 33$\times$ efficiency gain from the blessing of dimensionality alone.

A Background: Key Results from High-Dimensional Statistics

For reference, the main results we invoke:

Theorem A.1 (Matrix Bernstein inequality, Tropp 2012)**.** *Let $X_1, \ldots, X_M$ be independent random matrices with $\mathbb{E}[X_m] = 0$ and $\|X_m\| \leq L$. Then:*

$$\mathbb{P}\left(\left\|\sum_m X_m\right\| \geq t\right) \leq (d_1 + d_2) \exp\left(\frac{-t^2/2}{\sigma^2 + Lt/3}\right)$$

where $\sigma^2 = \max(\|\sum \mathbb{E}[X_m X_m^\top]\|, \|\sum \mathbb{E}[X_m^\top X_m]\|)$.

Theorem A.2 (Candès–Tao RIP, 2005)**.** *Let $\Phi \in \mathbb{R}^{M \times d}$ satisfy the restricted isometry property of order $2k$: for all $2k$-sparse x, $(1 - \delta_{2k})\|x\|^2 \leq \|\Phi x\|^2 \leq (1 + \delta_{2k})\|x\|^2$. If $\delta_{2k} < \sqrt{2} - 1$, then ℓ_1 minimization recovers any k-sparse signal exactly from $M = O(k \log(d/k))$ measurements.*

Theorem A.3 (Johnson–Lindenstrauss, 1984)**.** *For any $\epsilon > 0$ and n points in $\mathbb{R}^d$, there exists a linear map $\Phi : \mathbb{R}^d \to \mathbb{R}^m$ with $m = O(\log n/\epsilon^2)$ preserving all pairwise distances within factor $(1 \pm \epsilon)$.*